

IO500: The High- Performance Storage Community

Committee

- **Julian Kunkel** - Georg-August-Universität Göttingen/GWDG
- **Andreas Dilger** - The Lustre Collective
- **Dean Hildebrand** - Google
- **Jay Lofstead** - Sandia National Laboratories
- **Jean Luca Bez** - Lawrence Berkeley Lab

CONNECTING THE DOTS



ISC

High Performance

IO⁵⁰⁰

BoF Agenda

1. **Welcome** – Julian Kunkel
2. **Award Presentations** – Julian Kunkel
3. **Community Talk (15mins)** – Floyd Christofferson
4. **Updates**
 - New Community Policies – Andreas Dilger
 - Random Read Phase – Jay Lofstead
5. **Community Discussion** – Julian Kunkel

IO500 Organization Status

- A US non-profit, public charity organization: IO500 Foundation
 - Domain, mailing list, servers, GitHub belongs to IO500 Foundation
- Website contains results with links to details, CFS, BoF slides, etc.
 - io500.org
 - Contribute fixes at github.com/IO500/webpage
- Please join our mailing list for announcements:
 - io500.org/contact
- Please join our Slack for discussions:
 - io500workspace.slack.com/
 - Join link: rb.gy/sn8esm



Award Ceremony

IO⁵⁰⁰

10 Client Node Production - New Overall Winner

	# ↑	RELEASE	SYSTEM	INSTITUTION	FILESYSTEM TYPE	SCORE ↑	BW (GIB/S)	MD (KIOP/S)
New	1	ISC26	SCNet AICS-B	SCNet	ParaStor	7,839.30	2,551.40	24,086.69
	2	SC23	Aurora	Argonne National Laboratory	DAOS	2,885.57	734.50	11,336.27
	3	SC25	SuperMUC-NG-Phase2-EC-10	LRZ	DAOS	1,253.56	253.98	6,187.21
	4	SC25	Maximus-01	Core42	DAOS	666.70	247.90	1,793.03
	5	SC25	JoyBuilder AI Development Service Platform	JD Explore Academy	JPFS	405.50	117.37	1,400.93
	6	ISC25	HRT	Hudson River Trading	EXAScaler	348.08	136.05	890.51
	7	ISC24	Lise	Zuse Institute Berlin	DAOS	324.54	65.01	1,620.13
	8	SC24	GEFION	Danish Centre for AI innovation AS	EXAScaler	314.03	154.70	637.43
	9	SC25	CHIE-4	SoftBank Corp	EXAScaler	303.20	148.88	617.51
	10	SC24	CHIE-2	SoftBank Corp	EXAScaler	299.32	159.93	560.19
	11	SC25	HiPerGator	University of Florida	EXAScaler	273.84	79.39	944.56
	12	SC24	HiPerGator AI	University of Florida	EXAScaler	243.61	124.89	475.20

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Certificate

IO500 Performance Certification

This Certificate is awarded to:

SCNet AICS

Sugon ParaStor

#1 in the 10 Client Node Production Overall, Bandwidth and
Metadata Score

IO500



June 2026

IO500 Steering Board

<https://io500.org/list/ISC26/ten-production>

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10 Client Node Research - Two New Entries

	# ↑	RELEASE	SYSTEM	INSTITUTION	FILESYSTEM TYPE	SCORE ↑	BW	MD
							(GIB/S)	(KIOP/S)
	1	ISC23	Cheeloo-1 with OceanStor Pacific	JNIST and HUST PDSL	OceanFS2	137,100.00	2,439.37	7,705,448.04
	2	ISC23	Pengcheng Cloudbrain-II on Atlas 900	Pengcheng Laboratory	SuperFS	11,516.40	263.97	502,435.85
	3	SC22	ParaStor	Sugon Cloud Storage Laboratory	ParaStor	8,726.42	718.11	106,042.93
	4	SC22	StarStor	SuPro Storteck	StarStor	6,751.75	515.15	88,491.65
	5	SC22	SuperStore	Tsinghua Storage Research Group	SuperFS	5,517.73	179.60	169,515.95
New	6	ISC26	holmesai limited	holmesai limited	StorMeshAI	4,242.70	132.91	135,430.02
	7	SC23	Aurora	Argonne National Laboratory	DAOS	3,748.85	934.00	15,046.98
	8	ISC22	Shanhe	National Supercomputing Center in Jinan	flashfs	3,534.42	207.79	60,119.50
	9	ISC26	UCloud HPC Cluster	UCloud	UPFS2	2,638.18	71.07	97,935.96
	10	SC21	Athena	Huawei HPDA Lab	OceanFS	2,395.03	314.56	18,235.71
	11	SC21	OceanStor Pacific	Olympus Lab	OceanFS	2,298.69	317.07	16,664.88
	12	SC25	SuperMUC-NG-Phase2-10	LRZ	DAOS	1,997.41	323.11	12,347.70

IO500 Production List - New Overall Winner

	# ↑	RELEASE	SYSTEM	INSTITUTION	FILESYSTEM TYPE	SCORE ↑	BW (GiB/s)	MD (KIOP/s)
New	1	ISC26	SCNet AICS-A	SCNet	ParaStor	79,110.10	26,888.39	232,754.76
	2	SC23	Aurora	Argonne National Laboratory	DAOS	32,165.90	10,066.09	102,785.41
	3	SC25	SuperMUC-NG-Phase2-EC	LRZ	DAOS	3,470.22	861.22	13,982.88
	4	ISC25	Helma	Erlangen National High Performance Computing Center	Lustre	838.99	438.62	1,604.84
	5	ISC25	SSC-24	Samsung Electronics	WekaIO	826.86	248.67	2,749.41
	6	SC23	Shaheen III	King Abdullah University of Science and Technology	Lustre	797.04	709.52	895.35
	7	SC25	Maximus-01	Core42	DAOS	666.70	247.90	1,793.03
	8	SC24	IRIS	MSKCC	WekaIO	665.49	252.54	1,753.69
	9	ISC23	Leonardo	EuroHPC-CINECA	EXAScaler	648.96	807.12	521.79
	10	SC25	CHIE-4	SoftBank Corp	EXAScaler	551.85	399.41	762.47
	11	SC24	CHIE-3	SoftBank Corp	EXAScaler	500.20	331.66	754.41
	12	SC25	JoyBuilder AI Development Service Platform	JD Explore Academy	JPFS	405.50	117.37	1,400.93

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<https://io500.org/list/ISC26/production>

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IO500 Research List - Bandwidth

Sorted by BW

	#	RELEASE	SYSTEM	INSTITUTION	FILESYSTEM TYPE	SCORE	BW ↑	MD
							(GIB/S)	(KIOP/S)
New	1	SC23	Aurora	Argonne National Laboratory	DAOS		11,362.27	
	2	ISC26	Pengcheng CloudBrain with OceanStor A800	Pengcheng Laboratory	OceanFS		8,291.11	
	3	ISC23	Pengcheng Cloudbrain-II on Atlas 900	Pengcheng Laboratory	SuperFS		4,847.48	
	4	ISC23	Cheeloo-1 with OceanStor Pacific	JNIST and HUST PDSL	OceanFS2		2,439.37	
	5	SC25	SuperMUC-NG-Phase2	LRZ	DAOS		1,303.25	
	6	ISC23	Leonardo	EuroHPC-CINECA	EXAScaler		807.12	
	7	SC22	ParaStor	Sugon Cloud Storage Laboratory	ParaStor		718.11	
	8	SC23	Shaheen III	King Abdullah University of Science and Technology	Lustre		709.52	
	9	SC20	Oakforest-PACS	JCAHPC	IME		697.20	
	10	ISC20	NURION	Korea Institute of Science and Technology Information (KISTI)	IME		515.59	
	11	SC22	StarStor	SuPro Storteck	StarStor		515.15	
	12	SC23	BlueStorm	IBM Cloud HPC	IBM Storage Scale		440.99	

IO500 Research List - New Overall Winner

	# ↑	RELEASE	SYSTEM	INSTITUTION	FILESYSTEM TYPE	SCORE ↑	BW (GiB/s)	MD (KiOP/s)
New	1	ISC26	Pengcheng CloudBrain with OceanStor A800	Pengcheng Laboratory	OceanFS	603,335.00	8,291.11	43,903,983.64
	2	ISC23	Pengcheng Cloudbrain-II on Atlas 900	Pengcheng Laboratory	SuperFS	210,255.00	4,847.48	9,119,612.35
	3	ISC23	Cheeloo-1 with OceanStor Pacific	JNIST and HUST PDSL	OceanFS2	137,100.00	2,439.37	7,705,448.04
	4	SC23	Aurora	Argonne National Laboratory	DAOS	43,218.80	11,362.27	164,391.73
	5	SC22	ParaStor	Sugon Cloud Storage Laboratory	ParaStor	8,726.42	718.11	106,042.93
	6	SC22	StarStor	SuPro Storteck	StarStor	6,751.75	515.15	88,491.65
	7	SC25	SuperMUC-NG-Phase2	LRZ	DAOS	6,308.87	1,303.25	30,540.36
	8	SC22	SuperStore	Tsinghua Storage Research Group	SuperFS	5,517.73	179.60	169,515.95
New	9	ISC26	holmesai limited	holmesai limited	StorMeshAI	4,242.70	132.91	135,430.02
	10	ISC22	Shanhe	National Supercomputing Center in Jinan	flashfs	3,534.42	207.79	60,119.50
	11	SC22	HPC-OCI	Cloudam HPC on OCI	BurstFS	3,033.03	278.48	33,033.54
New	12	ISC26	UCloud HPC Cluster	UCloud	UPFS2	2,638.18	71.07	97,935.96

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IO500 Performance Certification

This Certificate is awarded to:

Pengcheng Laboratory
Cloudbrain with OceanFS
#1 in the IO500 Research Overall Score

IO500



June 2026

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<https://io500.org/list/ISC26/io500>

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List of Awarded Systems in the Ranked Lists

10 Client Production	Bandwidth Metadata Overall	SCNet AICS	ParaStor	2,551.40 24,086.69 7,839.30	GB/s KIOPS/s score
10 Client Research	Bandwidth Metadata Overall	JNIST and HUST PDSL	OceanFS2	2,439.37 7,705,448.04 137,100.00	GB/s KIOPS/s score
Production	Bandwidth Metadata Overall	SCNet AICS	ParaStor	26,888.39 232,754.76 79,110.10	GB/s KIOPS/s score
Research	Bandwidth Metadata Overall	Argonne National Laboratory Pengcheng Laboratory Pengcheng Laboratory	DAOS OceanFS OceanFS	11,362.27 43,903,983.64 603,335.00	GB/s KIOPS/s score

Community Talk

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A Case for Open Standards:

Addressing the Problem of Distributed Data
for Modern HPC/AI Workloads.

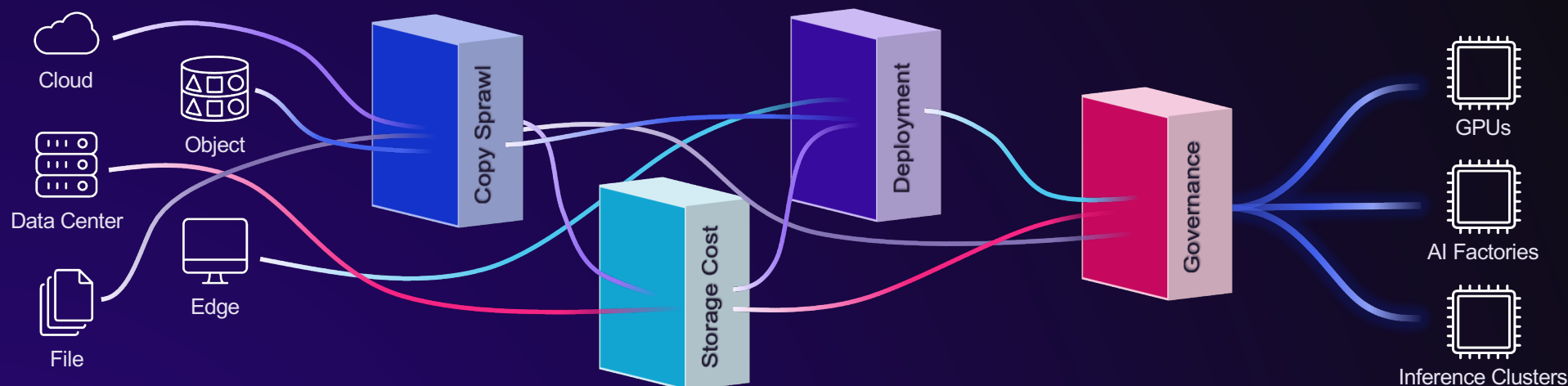
Floyd Christofferson
Hammerspace

ISC 2026 – Hamburg

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AI Exposes Limits of Centralized Data

AI is not just compute-constrained. It is data-constrained.



AI Changes How Data Must be Used

- Data stays where it is
- Compute moves to where it's available
- Workloads are no longer fixed

What Breaks At Scale

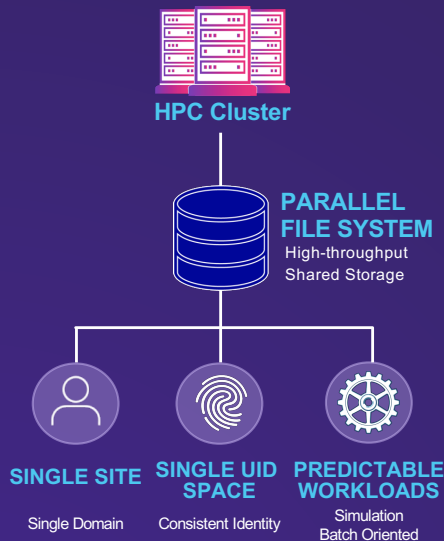
- Copy sprawl
- Idle GPUs
- Power & cost waste
- Deployment delays

The Reality Has Changed

From HPC Built for Mod/Sim to AI + Hybrid Workflows

Traditional HPC Was Built for Mod/Sim

Compute-Intensive. Predictable. Centralized.

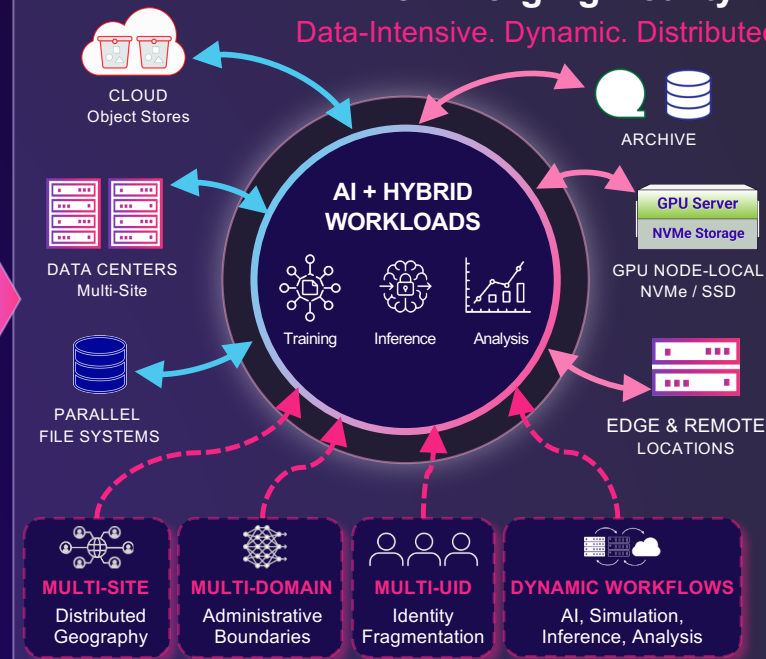


The Model That Worked

- ✓ Centralized storage
- ✓ Consistent identity
- ✓ Stable access patterns
- ✓ High-throughput shared file systems
- ✓ Built for long-running jobs

The Emerging Reality

Data-Intensive. Dynamic. Distributed.



What's Different Now

- ✗ Data is everywhere, not just in one place
- ✗ Identities don't always match
- ✗ Workloads are dynamic and diverse
- ✗ GPUs need data fast and local
- ✗ Moving or copying creates friction, cost, and risk



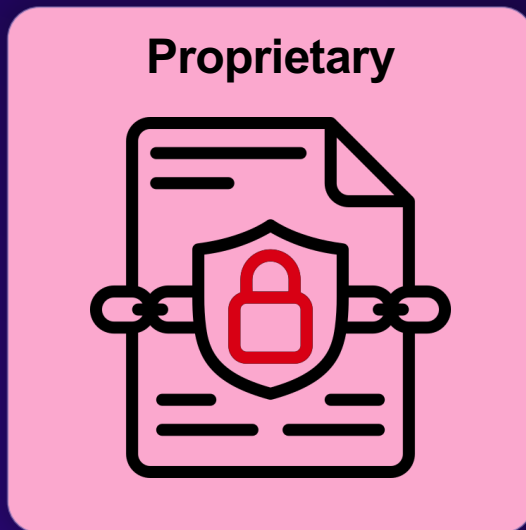
This is not just a Performance Problem.

It Is a Data Orchestration Problem

AI Requires Overcoming Proprietary Silos

AI is Driving Enterprise Adoption of
HPC and Web Infrastructure Architectures

From:



To:



Standards-Based Parallel File System Architecture

-- Parallel NFS v4.2 with Flex files --

Powering High-Performance:

- AI → Meta's Llama 2, 3, & 4
- HPC → Los Alamos National Lab
- VFX → Netflix/Animal Logic
- Web → PayPal

Hardening pNFS for HPC/AI Workloads

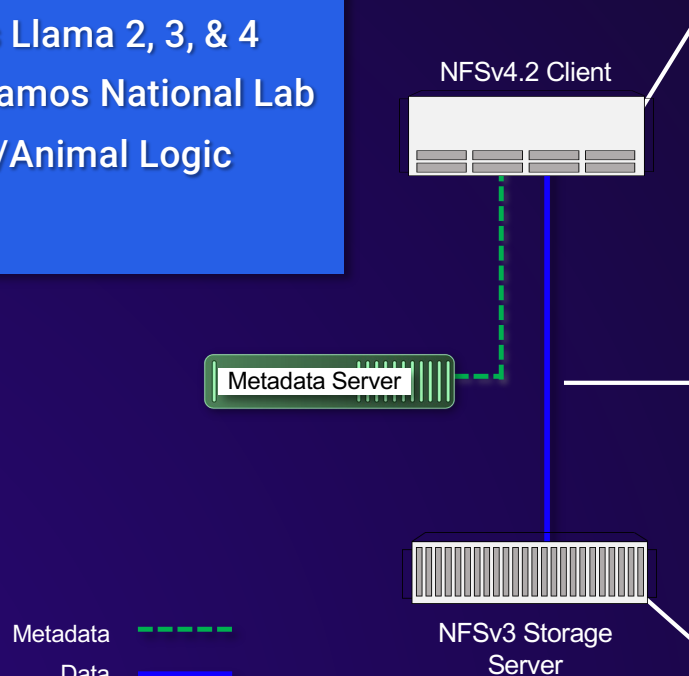
- 2018: Enhanced Parallel NFS spec with pNFSv4.2 with Flex Files
 - Eliminated NFS GETATTR chattiness
 - Added telemetry feedback
 - Added N-Connect
- Part of every modern Linux distribution

Separate Metadata and Data Paths

- Direct data path using TCP or RDMA
- Provides for multiple parallel connections
- Metadata layer acts as global control plane

Use Any NFSv3 Storage System

- Add any NFS storage volume, from any vendor
- Including GPU server local NVMe, or Tier 0



10 Node Production SC25 List

Customize

Download

Production

10 Node Production

Research

10 Node Research

Full

Ranking of production system submissions that used exactly ten client nodes. This is a subset of the Full List of submissions, showing only one highest-scoring result per storage system. Submitters who want a submission that is currently on the Research List to be on the Production List should contact the IO500 Steering Committee.

# ↑	INFORMATION							IO500		REPRO.	
	BOF	INSTITUTION	SYSTEM	STORAGE VENDOR	FILE SYSTEM TYPE	CLIENT NODES	TOTAL CLIENT PROC.	SCORE ↑	BW (GIB/S)		MD (KIOP/S)
1	SC23	Argonne National Laboratory	Aurora	Intel	DAOS	10	2,080	2,885.57	734.50	11,336.27	🟢
2	SC25	LRZ	SuperMUC-NG-Phase2-EC-10	Lenovo	DAOS	10	1,120	1,253.56	253.98	6,187.21	🟢
3	SC25	Core42	Maximus-01	Enakta Labs	DAOS	10	960	666.70	247.90	1,793.03	🟢
4	SC25	JD Explore Academy	JoyBuilder AI Development Service Platform	JDCloud	JPFS	10	960	405.50	117.37	1,400.93	🟢
5	ISC25	Hudson River Trading	HRT	DDN	EXAScaler	10	1,600	348.08	136.05	890.51	🟡
6	ISC24	Zuse Institute Berlin	Lise	Megware	DAOS	10	960	324.54	65.01	1,620.13	🟢
7	SC24	Danish Centre for AI innovation AS	GEFION	DDN	EXAScaler	10	1,280	314.03	154.70	637.43	🟢
8	SC25	SoftBank Corp	CHIE-4	DDN	EXAScaler	10	1,120	303.20	148.88	617.51	🟢
9	SC24	SoftBank Corp	CHIE-2	DDN	EXAScaler	10	1,120	299.32	159.93	560.19	🟢
10	SC25	University of Florida	HiPerGator	DDN	EXAScaler	10	1,920	273.84	79.39	944.56	🟢
11	SC24	University of Florida	HiPerGator AI	DDN	EXAScaler	10	640	243.61	124.89	475.20	🟢
12	ISC25	Joint Center for Advanced High Performance Computing	Miyabi-G	DDN	Lustre	10	640	188.26	77.38	458.06	🟢
13	ISC25	SAKURA Internet Inc and Prunus Solutions Inc	SAKURAONE	DDN	EXAScaler	10	640	188.26	77.38	458.06	🟢
14	ISC24	NHN Cloud Corporation	NHN CLOUD GWANGJU AI	DDN	EXAScaler	18	SC25	Samsung Electronics	Samsung AE-2 Cluster		
15	SC25	Simon Fraser University	FIR	DDN	EXAScaler	10	320	101.88	48.19	215.38	🟢
16	ISC24	University of Virginia	Afton	Dell	WEKA	10	850	105.94	33.28	337.29	🟢
17	SC23	Japan Agency for Marine-Earth Science and Technology	Earth Simulator 4	DDN	EXAScaler	10	320	101.88	48.19	215.38	🟢
18	SC25	Samsung Electronics	Samsung AE-25 Cluster	Hammerspace	pNFS v5.2.3-90 via Hammerspace Anvil Metadata Service	10	480	85.23	74.66	97.29	🟢
19	SC24	Poznan Supercomputing and Networking Center	Proxima	xFusion	Lustre	10	300	77.76	17.94	337.05	🟢
20	SC25	Purdue University	Gautschi	DDN	EXAScaler	10	1,920	70.18	19.59	251.47	🟢

The Result:

Top-20 IO500 Production Performance Using Standard Linux and pNFS

18

SC25 Samsung Electronics

Samsung AE-25 Cluster

Hammerspace

pNFS v5.2.3-90 via Hammerspace Anvil Metadata Service

10

480

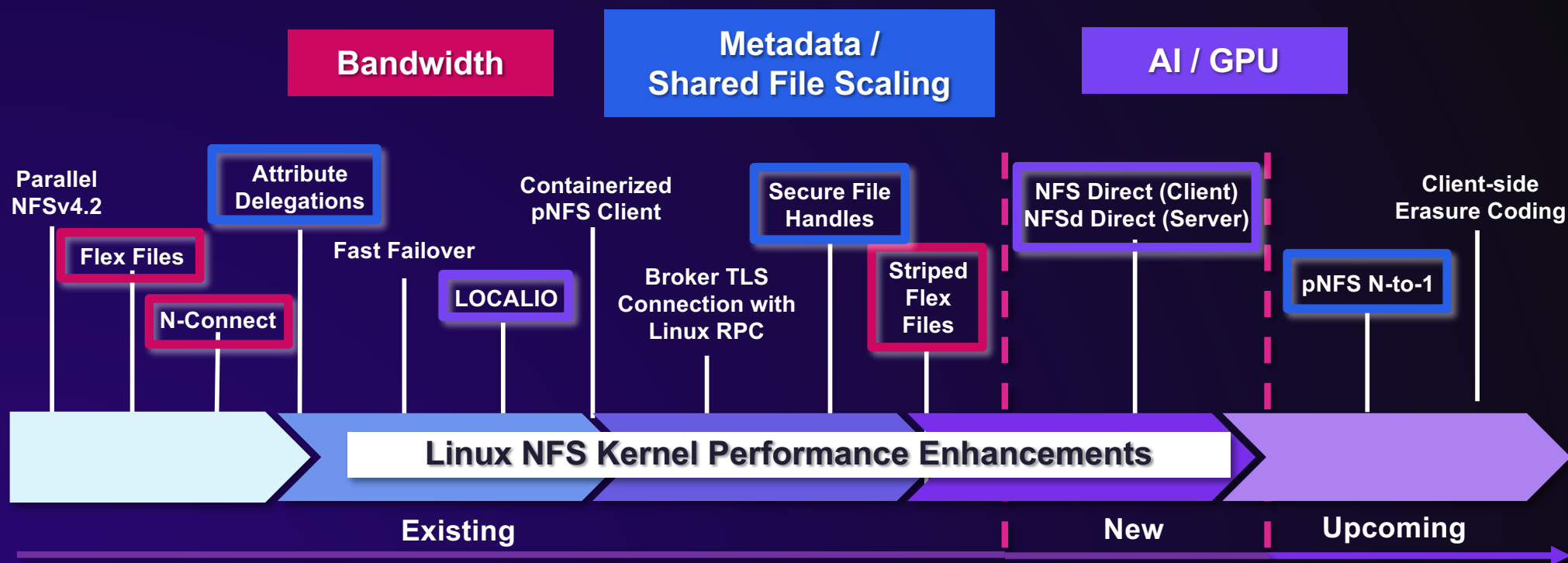
85.23

74.66

97.29

✓

Building Performance into Standard Linux



This Means Customers Never Have to Install Proprietary Client Software
Or Alter Existing Storage Servers from Any Vendor

Mapping Linux Enhancements to Real Workloads

Linux Enhancement	Primary Impact
Flex Files	Large sequential I/O
N-Connect	Client parallelism
Attribute Delegations	Metadata lookup scaling
Fast Failover	Production resiliency
LOCALIO	GPU-local data access
Striped Flex Files	Shared-file write workloads
NFS Direct	Direct storage access
pNFS N-to-1	Shared-file scaling

These weren't developed specifically to improve benchmark scores.

They were developed to solve real AI and HPC bottlenecks that happen to be measured by IO500.

Where pNFS Performs Well Today, And Where It's Still Improving

Strengths

- Large sequential AI workloads
- Metadata lookup and search
- Read-heavy pipelines

Focus areas

- Shared-file write contention
- Metadata-heavy checkpoint workloads
- Traditional Mod/Sim patterns

IO500 Phase	Result	Observation
IOEasy Write	509 GiB/s	Strong
IOEasy Read	541 GiB/s	Strong
IOHard Read	98 GiB/s	Improving rapidly
IOHard Write	1.15 GiB/s	Still a challenge
Find	488 kIOPS	Strong
MD Easy Stat	555 kIOPS	Strong
MD Hard Write	14 kIOPS	Area for future work

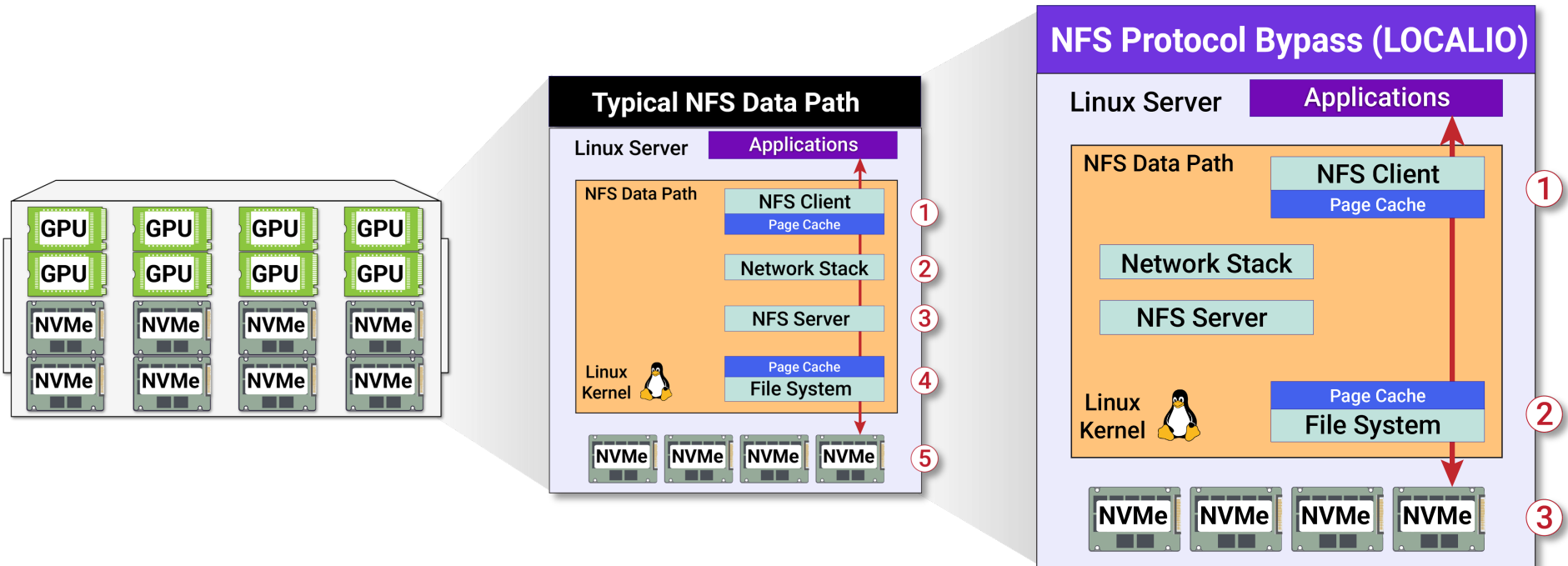
Can pNFS Match Traditional HPC File Systems?

Workload	Today	Direction
Large sequential AI training	Competitive	Strong
Large-scale inference pipelines	Competitive	Strong
Metadata search and discovery	Competitive	Strong
Shared-file write workloads	Improving	Active work
Checkpoint-heavy HPC workflows	Improving	Active work
Extreme metadata fanout	Still behind best specialized systems	Future work

- For many AI workloads, pNFS is already competitive.
- For some traditional Mod/Sim metadata workloads, specialized file systems still maintain advantages.
- The Linux community continues to work on those gaps.

Performance Benefits Waiting in the Linux Kernel

Further Reduce Latencies and Maximize GPU Utilization Using LOCALIO (NFS Protocol Bypass)

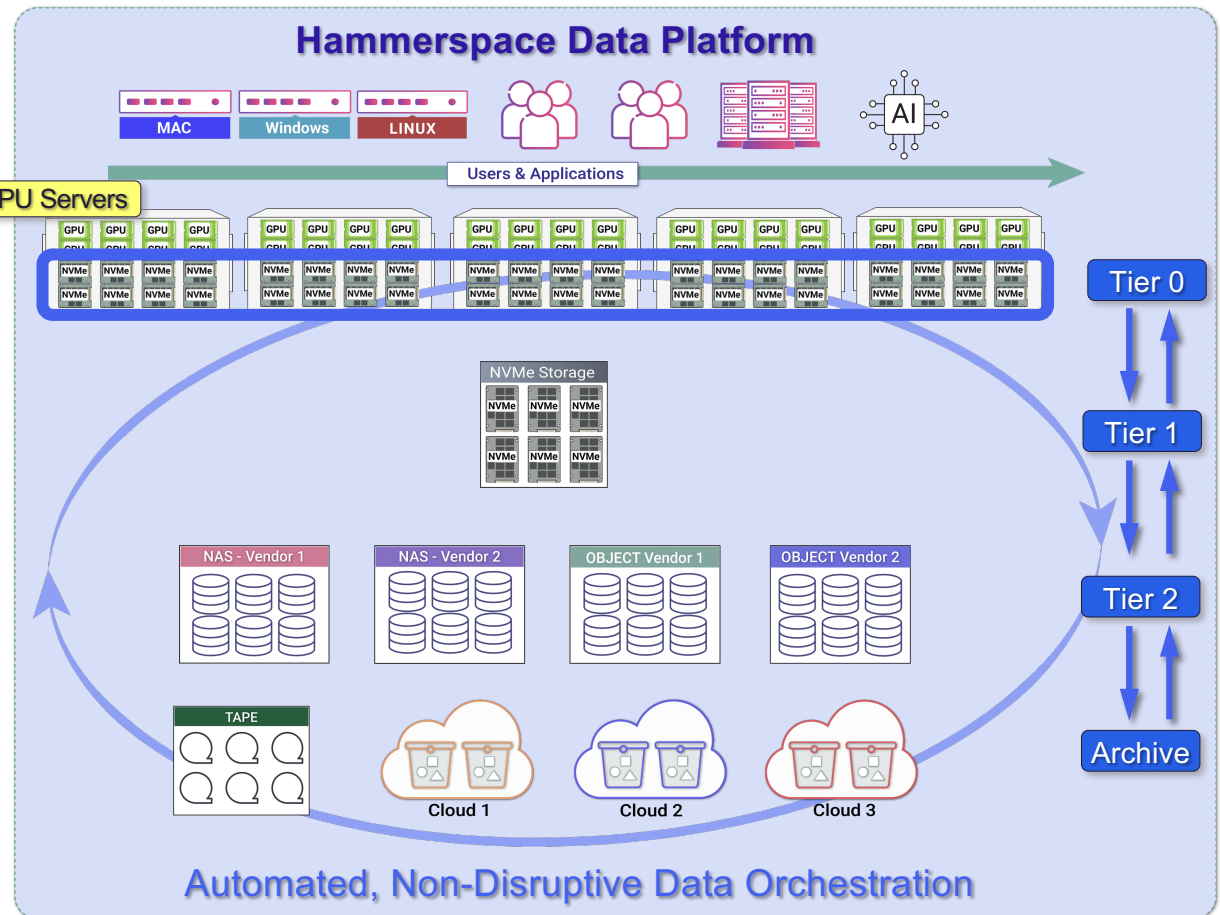


LOCALIO Was Released in Linux Long-Term Support kernel 6.12
Was included in RHEL10 in May 2025

Applying Standards-Based Performance to Tier 0

One example:
Using pNFS to activate
GPU-local NVMe
as a shared performance tier.

- Unifies local Server Storage with all other storage tiers
- Ensures all data is protected
- Reduces load on expensive Tier 1
- Eliminates vendor silos
- Automates AI workloads and data placement into & out of any other storage
- Spans multi-site & cloud



Why Standards Matter Beyond Benchmarks

Non-Proprietary, Standards-based Access

REST API

pNFS with Flex Files

S3, NFS, SMB

MCP CSI

Unified Data Plane

High-Performance Parallel Global File System – Cross-Platform Global Namespace

Automated Data Services

Service-Level Objectives for Data Placement, Workflows, Data Protection, Custom Metadata Services

Accelerated Data Orchestration

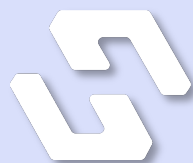
Non-Disruptive Data Movement for High-Performance Uses, Tiering, Replication, Cloud, Multisite, etc.

Silo-less Data Storage

Tier 0, pNFSv4.2 with Flex Files, Optimized Object Storage, Cloud Storage



Multi-Vendor, Multi-Site, Multi-Cloud Storage



**Hammerspace
Data Platform**

Why Open Standards Matter

- Standard Linux client
- Standard Linux server
- Standard pNFS protocol
- Top 20 IO500 Production result
- Ongoing Linux kernel innovation
- Applicable to AI and HPC Workloads

Thank You!

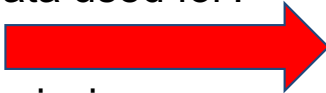
Come see us in Booth #A20

Floyd Christofferson – fc@Hammerspace.com

Policy Updates

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Policy Updates

- Shared 3 new policy proposals on April 15th on Slack (will ratify in July, 2026)
- Creation of new “Community Policies”
 - Govern behavior within the IO500 community
 - Submissions now require submitter to explicitly agree to these policies
- One small addition to IO500 Steering Committee Rules
 - Clarify submission data privacy - what is the data used for?
 - [Submission Data Handling Policy](#) Proposal 
 - Ownership and retention policies for submissions
 - Will not share contact info
 - User has right to request deletion, etc.
 - IO500 has permission to use data in perpetuity
 - Submitted data is also available to the public

Community Policy: Code of Conduct

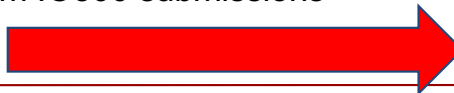
- Thankfully the IO500 community has been extremely well mannered
 - Want to formalize policies “just in case”
- Encourage behaviors that strengthen our community
 - Practice empathy, Respect diversity, Constructive feedback, Accountability, Community-first mindset
- Reminder that the following actions are strictly prohibited
 - Offensive content, Hostility, Harassment, Privacy violation, Unprofessionalism
- Applies within all IO500 community spaces
- Forward any violation to IO500 Steering Committee
 - Will try for prompt investigation and resolution
 - committee@io500.org
- [Code of Conduct Proposal](#) - QR code for URL 



Community Policy: Messaging Policy

- Govern how submitters can use and reference IO500 List information and results
- Formalize “Verified” and “Unverified” results
 - **Verified** - Submission is on a published IO500 List
 - **Unverified** - IO500 benchmark score not on any IO500 list
- All public messaging must prominently state Verified or Unverified
- Specify how data can be characterized, for example
 - Must state list instance, cannot directly compare Production vs. Research results
 - Must state actual list ranking position(s), use metrics from actual lists
- IO500 Steering Committee handle violations by:
 - Direct Outreach for preferably quick correction/resolution
 - If submitter does not quickly resolve issue, increasing actions taken:
 - Issue a formal statement
 - Submission Removal
 - Organizational Sanction and even Ban from IO500 submissions

[Messaging Policy Proposal](#) - URL in QR code



Website Updates

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Website Updates

- Submission System:
 - Improved input field descriptions
 - Additional options on selection fields (e.g. interconnect, architecture, and file system)
 - Reach out if you noticed something missing!
 - Improved security of stored information
 - Adhering to the updated Community Data Retention Policy
 - Working to address raised issues:
 - Complexity of submission form
 - Validation of minimum set of required fields
- Website:
 - Improved field filtering available when building lists
 - Fixed random-read display and added GiB/s alongside kIOP/s
 - Migrated plots to dynamically update upon list releases
 - Fixed broken download links for previous submissions files

Benchmark Phases and Extended Access Patterns

4K Random Read Phase

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Latest IO500 Benchmark Updates

- New Phases
 - Added ior-rnd4k-easy-read phase (discussed in next section)
 - Still trying to define a find-hard phase
 - Add a directory rename phase?
 - Planning to remove all previous optional phases
- Scoring Adjustment?
 - Metadata scores growing very large and overshadowing bandwidth due to “find-easy”
 - Considering rebasing metadata scores from KIOPs to MIOPs, but affects rankings
- List Download
 - Some fields missing, working to improve submission form to ensure completion
 - Per-FS fields makes comparisons difficult, try to map to a common flat schema
- **io500.org** Submissions page
 - Please continue to give feedback

ior-rnd4k-easy-read Phase - Current Status

- Random 4KB read phase first added in SC24 (unranked)
 - Reuse existing ior-easy-write files for input to avoid writing new files
 - Total data size is the largest available from previous phases to minimize caching
 - Run at end of other phases to avoid conflicting with other phases/scores
- Existing score is kept, add new score with ior-rnd4K-easy-read phase
- Current status of benchmark results
 - ISC25+ - phase run by default (unless disabled), result is not part of official score
 - ISC26 - **issue if write very slow & large RAM, files too small to exceed RAM**
 - May need to invalidate some older ior-rnd4k results due to cache effects
 - Limit runtime to 60s stonewall or $\frac{1}{3}$ of total file data read to minimize cache hits
 - Change PRNG to use LFSR for fully non-repeating block read sequence
- Next steps to include it into benchmark runs/score
 - Propose when 6+ of Top-10 list entries have new score trigger move to new ranking
 - Use 1% of ior-easy-read score for old submissions (low end of actual scores)

Current ior-rnd4k-easy-read status

- 4KB blocks
 - 1 MB/sec * 300 seconds = 76,800 blocks (17 bits)
 - 200 GB/sec * 300 seconds = 15,728,640,000 blocks (34 bits)
 - 2 400 Gb/sec NIC = 100 GB/sec
- PRNGs evaluated
 - Mersene Twister, Philox, Xoroshiro, LCG (rand_r)
 - Linear-Feedback Shift Register (uniform distribution across full period)
- Approach:
 - Loop until x% of file covered
 - ceiling (power of 2) of remaining file size
 - LFSR walk of all 4K blocks

Voice of the Community & Open Discussion

IO⁵⁰⁰

Other Potential Access Patterns

- Should an `ior-rnd4k-write` phase also be added?
 - Relatively few HPC workloads have purely random writes
- Want `find-hard` to be “harder” than “find in `mdtest-hard/` directory”
 - Existing `find` score is totally unbalancing the other metadata results
 - Save full list of filename(s) into a file in the storage system instead of just count
 - Extra attributes, something other than filename (string) comparison?
 - Variable parameters to avoid pre-generating the result during file creation
 - Geometric mean of `find-hard` and `find-easy` to replace existing `find`?
- Should other random read/write phases be added?
- Should a directory rename phase be added?
- Expect runtime would increase by about 20 min

SC 26 (Nov 15-20, 2026)

- Call for submission: Oct 1st
- Submission deadline: Nov 10th
- List release: BoF date TBD



Open Floor

- Downloading and comparing submissions
 - Collecting storage system metadata automatically
 - Is the submission form getting better?
 - How to make 'find-hard' really hard
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- Community Survey on various topics - QR code

